

AXIOM

Zero-Trust Invariant & Blast-Radius Firewall for Autonomous AI Dev Partners

Event: IBM Bob 2.0 Hackathon on lablab.ai | **Author:** Anmol Suwalka

Repository: github.com/suwalkaanmol/axiom

Target Problem: AI dev tools write code 10x faster, but introduce silent unstated assumptions that bypass human review.	Core Innovation: Adversarial counter-agent that mines ghost invariants, attacks them with fuzzing, and guides IBM Bob self-healing.	Enterprise Value: SOC2/in-toto cryptographic Resilience Passports ensuring safety before deployment.
---	---	--

1. The Enterprise Crisis: The AI Assumption Blindspot

When AI dev partners (like IBM Bob) refactor multi-file repositories, code looks clean and passes happy-path tests. However, AI code harbors dangerous unstated assumptions that cause catastrophic production outages.

1. Boundary Neglect	2. Tenant Leakage	3. Concurrency Races	4. Idempotency Gap
AI assumes numbers/quantities are always positive (>0). Negative amounts cause infinite balance duplication exploits.	AI omits organizational workspace checks, allowing cross-tenant data/fund contamination between clients.	Non-atomic read-modify-write mutations collapse under burst traffic, resulting in double-spending.	Missing replay token deduplication causes network retries to execute financial transactions repeatedly.

The Human Reviewer Paralysis: Reviewers cannot audit 500-line multi-file AI pull requests for implicit edge-case contracts. They rubber-stamp PRs, allowing fragile code into production.

2. The Solution: Zero-Trust Invariant & Blast-Radius Engine

Axiom acts as an autonomous adversarial auditor that sits directly in the CI/CD pipeline, breaking the circular confirmation bias of AI testing AI.

1. Invariant Mining	Inspects AST & semantic diff to extract unstated assumptions (boundary, tenant isolation, concurrency).	Exposes hidden assumptions before code execution.
2. Adversarial Fuzzing	Synthesizes property-based adversarial pytest suites to deliberately break extracted invariants.	Proves vulnerabilities on camera with reproducible stacktraces.
3. Blast Radius Map	Interactive system topology mapping downstream PostgreSQL databases, Stripe APIs, and Kafka streams.	Immediate visual risk assessment (Risk Score 88/100).
4. Self-Healing with Bob	Feeds failure stacktraces back to IBM Bob to synthesize zero-trust hardened patches.	Automated remediation loop (Risk drops to 12/100).
5. Resilience Passport	Issues in-toto / DSSE cryptographic attestation hash and verified GitHub PR badge.	Audit-ready compliance for enterprise SOC2 standards.

3. Live Demonstration: FinTech Wallet Transfer

Axiom in action against a real-world financial transaction service written by IBM Bob.

Step 1: AI Code Generated	Step 2: Adversarial Attack	Step 3: IBM Bob Remediates
IBM Bob generates transfer_funds(). Happy path unit tests pass (100% green). Axiom Alert: 4 Ghost Invariants flagged. Downstream PostgreSQL database marked as Critical Blast Radius (Risk: 88).	User clicks EXECUTE ADVERSARIAL FUZZ. Result: 3 FAILED / 0 PASSED. Terminal catches credit duplication exploit: passing amount = -500 artificially incremented sender wallet balance!	User clicks TRIGGER IBM BOB AUTO-PATCH. Bob enforces zero-trust bounds and tenant checks. Result: 3 PASSED / 0 FAILED. Topology turns green, Risk drops to 12/100.

Key Takeaway: Axiom provides definitive mathematical and empirical proof of code safety rather than superficial text summaries.

4. Technical Execution & Enterprise Value

Axiom was built from the ground up for the IBM ecosystem to solve the biggest bottleneck to autonomous enterprise adoption.

• Backend: Python 3.14, FastAPI, AST Analysis, pytest subprocess sandbox • Frontend: React 18, Tailwind CSS, Vite, dynamic topology node graph • AI Partner: IBM Bob 2.0 / watsonx for code synthesis and self-healing loop • Attestation: in-toto / DSSE cryptographic SHA-256 signature generator • Open Source: Public GitHub repository with 1-click launcher and verified unit tests	• Not a Toy Chatbot: 90% of competitors build chat assistants. Axiom solves the hard enterprise problem: automated code verification. • High Visual Wow-Factor: Interactive topology graph and live pytest execution terminal give judges a thrilling 2-minute demo. • IBM Synergy: Elevates IBM Bob from an experimental tool into a trusted, enterprise-ready dev partner. • Enterprise Compliance: Delivers audit-ready provenance and resilience passports for enterprise CTOs.

Live Repository: <https://github.com/suwalkaanmol/axiom> | **Developer:** Anmol Suwalka